Capture-Complete, Prediction-Incomplete Offline Reinforcement Learning for Transferable Bullet-Hell Control

Touhou Solver Research Project

Working draft — August 21, 2026

Abstract

We study an offline reinforcement-learning agent for the original Japanese Touhou 6 version 1.02h under Wine, targeting a complete Lunatic no-miss, no-bomb route and eventual transfer to Touhou 8. The central hypothesis is that the runtime need not predict every script instruction or future bullet birth. Instead, the environment adapter should capture complete, coherent physical facts; a small online shield should reject actions that collide with already instantiated hazards over a bounded horizon; and unknown pattern dynamics should be inferred from object-centric history by an offline learner. This draft specifies the system boundary, immutable data contract, falsifiable research questions, and preregistered experiments. It intentionally reports an audited canonical six-stage infrastructure baseline with 108 retained HITs and no infrastructure failure; it reports no learning result.

1 Motivation and Scope

Bullet-hell control combines dense continuous geometry, discrete input, partially observed pattern state, long episodes, and sparse but unambiguous failure events. A tempting engineering strategy is to reconstruct every game script opcode and predict every future hazard. That strategy couples the controller to one title, creates a large second implementation of the game, and makes failures difficult to attribute: a HIT may come from the controller, the learned policy, or the hand-written simulator.

TOUHOU-OFFLINE instead adopts a *capture-complete, prediction-incomplete* boundary. The adapter records what physically exists at each coherent root, while offline learning is responsible for regularities not encoded by the minimal shield. The immediate target is a complete Touhou 6 Lunatic route with HIT continuation and zero Bomb. The research target is Lunatic NMNB. A second title, Touhou 8, is reserved as a transfer test rather than a source of training-time identifiers.

This is a working protocol, not a result paper. In particular, we do not yet claim that the observed-hazard shield prevents all HITs, that object history is sufficient to infer unknown births, that offline RL improves the behavior policy, that accelerated Wine is evidence-equivalent, or that the method transfers to Touhou 8.

2 Research Questions

RQ1: minimum online semantics. Can a hazard first instantiated after a paused decision root damage the player before the next controllable root, and can its action-relative risk be learned from physical history without adding title-specific source forecasting online?

RQ2: learnability of unknown dynamics. Does player state plus a short history of unordered physical objects contain enough information to predict action-relative future HIT cost on held-out complete episodes?

RQ3: data longevity. Can the same immutable physical episode corpus train materially different offline algorithms without recollection and without source-script features?

RQ4: transfer. Can the episode schema, feature contract, learner, exporter, and evaluator remain unchanged when only the Touhou 6 memory/input adapter is replaced by a Touhou 8 adapter?

3 System Boundary

The system has three independently testable components.

1. A Wine adapter pauses the exact process and records a coherent physical root, input delivery, behavior probability, outcomes, lifecycle, and provenance.
2. A native shield filters actions whose bounded paths overlap hazards already instantiated at that root.
3. An offline learner consumes complete generic episodes and exports a frozen scorer. The scorer may rank the shield set but may never add an action to it.

There is no mandatory online ECL interpreter, stage route, spell rule, boss rule, frame window, coordinate target, or pattern exception. Extracted source scripts may support forensic analysis of RQ1, but source identities are not actor inputs and their successful decoding is not a corpus-admission condition. HIT is an observed cost and does not terminate data collection; Bomb is forbidden.

4 Minimal Online Shield

At decision root t, let x_t be player state, A the 18 directional/focus actions, and O_t the instantiated bullets, lasers, and enemy collision bodies. The adapter lowers O_t to per-frame hazard primitives for a fixed horizon $H = 4$. For each candidate $a \in A$, the native kernel advances the player under the measured movement constants and bounded input-delivery paths. It returns

$$A_{\text{shield}}(x_t, O_t) = \{a \in A : d(x^a_{t+k}, O_{t,k}) > m,\ k = 1, \ldots, H\}, \tag{1}$$

where $m = 0.35$ pixels is the fixed additional collision margin and d uses the source collision geometry.

The claim is deliberately narrow: a returned action is clear of the stored projection of hazards observed at root t. The shield makes no claim about a hazard born after t. The source-order audit at TH06 commit `cc475a0bc3fef38683b0f02224c87ddba0a021d9` establishes this boundary: Player update priority 7 precedes EnemyManager priority 9 and BulletManager priority 11; enemy execution may instantiate a hazard which BulletManager can test for collision in that same update. The audited definitions are in `src/ChainPriorities.hpp`, `src/EnemyManager.cpp`, and `src/BulletManager.cpp` of that source revision. Such births are ordinary partially observed learning outcomes, not justification for a title-specific online predictor. Empty shield sets are recorded as control dead ends; they are not silently widened.

The online path therefore contains only coherent capture, observed-object kinematics, native collision filtering, frozen-policy scoring, and input publication. A retired hand-written long-horizon beam planner is excluded so that gameplay changes can be attributed to the learned policy rather than an untracked rule system.

5 Algorithm-Independent Episode Data

An episode is an immutable sequence

$$\tau_i = \{(o_t, A_t^{\text{shield}}, a_t, \mu_t, o_{t+1}, c_t, b_t)\}_{t=0}^{T_i-1}, \tag{2}$$

where o_t contains factual physical state, a_t distinguishes proposed, published, sampled, and physically witnessed execution, $\mu_t(a)$ is the complete behavior distribution over the admissible actions, c_t contains physical HIT and resource deltas, and b_t records lifecycle boundaries and infrastructure exclusions. Frames and transitions are joined by immutable references and shard hashes. A complete episode has exactly one fewer transition than frame.

The canonical raw layer retains measured player state; occupied bullet, laser, enemy, player-shot, and item records; visual and collision geometry; resource counters; input state; clocks; capture retries; and executable, native, policy, and schema provenance. Derived tensors are versioned products rather than replacements for raw shards. This permits multiple feature builders and RL algorithms to consume identical episode identities.

Deployable features may contain normalized player state, unordered physical object primitives, action history, declared portable resources, and the shield mask. They may not contain stage, boss, spell, ECL, RNG, memory address, run identifier, future outcome, or post-treatment facts. Unordered hazard sets motivate permutation-invariant encoders such as Deep Sets [4], but this complexity is admitted only after a transparent vector baseline passes.

6 Offline Learning Ladder

Complexity is added only after the preceding rung produces a decisive signal.

1. Validate schema, linkage, propensity, HIT conservation, and forbidden feature leakage; report an action-frequency baseline.

2. Fit transparent behavior cloning (BC) on current portable observations,

$$\mathcal{L}_{\text{BC}}(\theta) = -\sum_{i,t} \log \pi_\theta(a_t \mid o_t, A_t^{\text{shield}}). \tag{3}$$

3. Add only a short factual observation/action history and measure the held-out improvement as an ablation.

4. If needed, add a permutation-invariant object encoder and then a small temporal model.

5. Only after representation learnability, compare BC with the smallest justified offline value learner. IQL is one candidate because its critic update avoids direct evaluation of out-of-dataset actions and its actor is extracted with advantage-weighted BC [2]. CQL is a separate conservative baseline [3]; neither is assumed superior in advance.

6. Add uncertainty abstention or auxiliary factual dynamics objectives only as separately measured ablations.

Sequence-modeling methods such as Decision Transformer [1] are explicitly deferred. A larger model does not repair a broken capture contract, leakage, unsupported actions, or episode-level train/test contamination. The primary cost is physical HIT count with undiscounted episodic accounting; auxiliary predictions never change reward.

7 Experimental Protocol

Every result must name the Git commit, executable hash, policy hash, schema, full command, complete episode IDs, and artifact paths. Splits are by complete episode. Negative results are retained. Infrastructure failures are reported separately and never counted as gameplay improvements.

ID	Decisive question	Status
E0	Does the reset runner complete Practice 4–6 and one full route?	Completed
E1	Can a post-root birth hit before the next controllable root?	In progress
E2	Do generic loaders preserve facts without ECL streams?	In progress
E3	Do portable BC features beat action frequency out of episode?	Future work
E4	Does offline policy improvement reduce paired route HIT cost?	Future work
E5	Are parallel/accelerated collectors factually equivalent?	Future work
E6	Which surfaces change under a Touhou 8 adapter?	Future work

Table 1: Preregistered experiment ledger. Status changes require immutable artifacts.

7.1 E0: infrastructure baseline

Run natural Lunatic Practice Stages 4, 5, and 6 and then one natural six-stage route on original Wine at normal timing with the frozen baseline policy. HIT continuation and zero Bomb are mandatory. Accept only if every declared run completes, frames equal transitions plus one, no record is dropped, player successors and stored shield decisions replay exactly, every HIT has linked before/action/after evidence, and cleanup leaves no worker process. HIT count is measured, not gated.

At solver commit `af9900524520b72934a4c55e2f44118f88094633`, all three bounded Practice runs completed under isolated four-CPU workers after the atomic-input cleanup. Table 2 reports their complete audits. Every declared frame and transition loaded, every Bomb/callback/integrity/scope count was zero, all checked player successors were bit-exact, and every stored dense shield sample reproduced with no unsafe or conservative divergence. The higher Practice observation-gap rates motivated eight CPUs for the canonical route and future evidence collection; gap transitions remain explicit and are excluded rather than fabricated.

The Stage 4–6 run IDs are, respectively, `20260821T080408Z-471737400`, `20260821T081924Z-372154700`, and `20260821T080532Z-100425500`; their audit SHA-256 digests are `c4178c3c9e5b7d88f7a950d8fe18403630a8524320ae72dc0d0ce97c0a26bb2d`, `2677f2b9689c42c5483a2e278d077a6b721623430d89f6c39a41bc92b3c34ebd`, and `e34bec2844dc7ce92f572326dd262a1e7f718d6798bc9a6c9dd7124692308c17`.

The natural-RNG full route then reached Ending with run ID `20260821T083303Z-499458600`: 135,561 frames, 135,560 transitions, 108 physical HITs, zero Bomb, zero dropped record, and zero capture, corpus, policy, trace, or other infrastructure failure. All six Stages were present. The

Stage	Frames	Transitions	HIT	Gap %	Successors	Shield
4	25,564	25,563	18	0.536	24,344	58
5	23,129	23,128	23	0.968	19,503	64
6	30,981	30,980	34	0.507	28,763	64

Table 2: Canonical E0 Practice evidence. Successor and shield columns are checked counts; every mismatch/divergence count is zero.

audit classified all 108 HITs as observed-shield-empty, reproduced 64 dense shield samples without divergence, and matched 126,939 eligible player successors bit-for-bit. Only 28 links were excluded for observation gaps (0.0207%); capture and solve p99 were 8.044 and 0.918 ms respectively. The independent baseline verifier passed every declared check. The immutable SHA-256 digests for report, audit, run metadata, and manifest are respectively `f2577d31ca5a51108911b4d60a957a7e2e891edda5ba337fbca1f70386e37686`, `8d13d3196261f4f8710d8187cdec898e364b67f024696a8cca1991aff467baa9`, `9f9fbdd75dbe89651c789dc502dab6a50323a93ca488051720538308079e3396`, and `8423bae1ef93decd50605a5e47900ee0ded3a6096d6067fb4c0a8eca944ec1d8`. E0 is therefore complete; the 108-HIT value is a baseline measurement, not a policy-quality claim.

7.2 E1: immediate-birth boundary

Use source update order and adjacent physical roots around every HIT and newly observed overlapping object. Source order already proves that post-root birth and same-update collision are possible. The remaining experiment links this class to adjacent Wine roots and measures whether short physical history predicts its action-relative cost. No source/ECL envelope is added to the online shield; pattern-specific exceptions are forbidden.

7.3 E2–E4: data, learnability, and policy improvement

E2 removes optional source/ECL material and requires the generic loader to reproduce action sequences, HIT totals, episode grouping, and hashes while two different feature builders consume the same raw episodes. E3 compares action frequency, current-observation BC, and short-history BC using held-out negative log likelihood, accuracy, calibration, action support, and episode bootstrap intervals. E4 begins only if E3 passes. It compares BC against one offline value method at a time, with held-out factual HIT-cost prediction and action-relative discrimination offline, followed by paired complete-route HIT count and NMNB completion rate online.

7.4 E5–E6: collection equivalence and transfer

Parallel workers are enabled only after E0. Serial and parallel fixed-seed runs must first agree on factual/action traces before independent natural-RNG collection begins. The first declared collection policy is an immutable 20% mixture of uniform exploration over the observed-shield set and the generic reactive baseline, with complete propensities and independent policy RNG streams per episode. Clock acceleration is a distinct hypothesis and cannot produce evidence until it passes a normal-speed differential contract. E6 replaces only the environment adapter and reports unchanged versus changed code/schema surfaces, zero-shot behavior, and fine-tuning sample cost.

8 Threats to Validity

The shield horizon may be too short, and unknown births may violate its narrow claim. Behavior support may be insufficient for action-relative learning; complete-route episodes may be too few for reliable uncertainty estimates; Wine scheduling may introduce observation gaps; and Touhou 6–specific memory facts may leak through an apparently generic feature builder. E0 and E2 are therefore prerequisites, not housekeeping. The transfer claim is falsified if downstream learner or corpus semantics require title-specific branches.

9 Artifact and Reproducibility Contract

The implementation uses repository-relative scripts. The LaTeX artifact is built by `scripts/check_paper.sh`. Runtime setup, native compilation, Wine execution, corpus validation, and route verification have distinct entry points. The canonical compiled paper is tracked at `paper/main.pdf`; intermediate LaTeX files stay in the ignored `paper/build/` directory. Wine prefixes, corpora, and fitted models remain untracked outputs. Committed experiment claims must point to immutable hashes rather than mutable filenames.

10 Conclusion

The project asks whether correct physical capture plus a small observed-hazard shield is a better foundation for transferable bullet-hell offline RL than a title-complete hand-written simulator. The architecture is intentionally minimal so that the first full-route baseline can distinguish infrastructure failure from learning failure. The answer remains empirical: E0 establishes the data plane, E1 establishes the explicit partial-observation boundary, and only then do E3–E6 evaluate learning and transfer.

References

[1] Lili Chen, Kevin Lu, Aravind Rajeswaran, Kimin Lee, Aditya Grover, Michael Laskin, Pieter Abbeel, Aravind Srinivas, and Igor Mordatch. Decision transformer: Reinforcement learning via sequence modeling. In *Advances in Neural Information Processing Systems*, 2021. URL `https://arxiv.org/abs/2106.01345`.

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[3] Aviral Kumar, Aurick Zhou, George Tucker, and Sergey Levine. Conservative q-learning for offline reinforcement learning. In *Advances in Neural Information Processing Systems*, 2020. URL `https://arxiv.org/abs/2006.04779`.

[4] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabás Póczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, 2017. URL `https://arxiv.org/abs/1703.06114`.